

- A derived cluster specification MAY have mandatory requirements that are optional in the base specification.
- A derived cluster specification MAY remove requirements that are optional in the base specification.
- A derived cluster specification MAY remove or make optional a requirement that is mandatory in the base specification, if the resulting specification is deemed useful in its reduced form, and logically a subset of the base clusters.

For example: The Bridged Device Basic Information cluster is derived as a reduced form of the base Basic Information cluster, where many informational attributes are not mandatory, because the information is not available from devices behind the bridge. However, the derived cluster provides the same, but reduced, function as the base cluster.

- It is RECOMMENDED that a derived cluster makes changes to the base cluster by extending or reducing one or more features or independent elements, and not by modifying features, elements or cluster behavior.

Such modifications SHOULD be defined as a feature, which can be used to extend or reduce.

- All new features, elements or behavior introduced by the derived cluster that may be useful across other derivations SHOULD be defined in the base cluster specification and made optional (in that base cluster specification), to maintain the entire set of requirements and identifier namespace in one place.
- When a derived cluster is created, the base and derived cluster SHALL define a limited range of element ID values allocated for the base and derived cluster, to ensure no overlap will occur. A limited range allows for sufficient use of the ID space for most use cases while allowing additional ranges to be defined in the future should these ranges become exhausted. The range SHALL consider any existing element IDs or bits in the existing clusters, but if possible the following ranges are recommended as the default values reserved for a base cluster:
 - Feature bits: 0-19
 - Attribute IDs: 0x0000-0x007F
 - Command IDs: 0x00-0x7F
 - Event IDs: 0x00-0x7F

NOTE

Readers should be aware that the Color Control, Door Lock, Level Control and OnOff clusters do not follow these recommended ranges, due to being created before these recommended ranges were defined.

- A derived cluster specification SHALL define its own revision ([ClusterRevision attribute](#)) that is independent of the base specification, with the single exception that if the base specification is modified/updated in a way that requires its revision to be increased, as detailed in [Section 7.10.1, “Cluster Revision”](#), then the revision of any derived cluster specifications SHALL also be increased.
- A base cluster specification MAY be created from an original base cluster, which then becomes a